



Foundations of Finance



Capital Budgeting Rules



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Partial Review of the Last Class

- ▶ A share pays $D_1 = \$3$, trades at \$60 and its dividend grows 4% a year. What return does the market expect?
- ▶ A firm pays out 40% of its earnings and earns an ROE of 10%. How fast does it grow?
- ▶ ROE is 8% and r is 10%. Should the firm keep its earnings or pay them out?
- ▶ In a two-stage DDM, why is the Gordon price at year T discounted T periods and not $T + 1$?

Today

- ▶ From valuing a share to deciding on a project
- ▶ The NPV rule, and why it always works
- ▶ The IRR rule: a project's yield to maturity
- ▶ Three ways the IRR rule fails
- ▶ The payback rule, and what it ignores
- ▶ What managers actually use
- ▶ The sample midterms

From Shares to Projects

- ▶ Last week we asked what a share is worth. Today the firm asks what a project is worth
- ▶ **Same formula: $NPV = C_0 + \sum C_t / (1 + r)^t$**
 - ▶ a bond discounts promised coupons, a share discounts expected dividends, a project discounts expected cash flows
- ▶ Two inputs we take as given today:
 - ▶ the cash flows — how to build them from the accounts is Sessions 11 and 12
 - ▶ the discount rate r , the project's cost of capital — where it comes from is Session 20
- ▶ If a project is worth more than it costs, the firm — and its share price — is worth more

How Are These Decisions Made?

Company	Investment Decision
Intel (U.S.)	Invests \$7 billion in expanding semiconductor plant in Chandler, Arizona
Amazon (U.S.)	Acquires self-driving startup, Zoox, for over \$1.2 billion
Tesla (U.S.)	Announces construction of new plant to build the electric Cybertruck
Shell (U.K.)	Starts production at a deepwater development in the Gulf of Mexico
BYD (China)	Increases production capacity to meet fast-growing demand for its electric cars
Ørsted (Denmark)	Completes a 230-MW wind farm in Nebraska
Unilever (U.K.)	Spends \$8 billion on advertising and marketing
Ambev SA (Brazil)	Spends \$268 million on R&D
GlaxoSmithKline (U.K.)	Acquires Canadian biopharmaceutical company Bellus Health for approximately \$2.0 billion
Reliance Industries (India)	Announces plans to establish a fully integrated solar photovoltaic gigafactory in India

Capital Budgeting Decisions

- ▶ **Examples of capital budgeting decisions made by firms:**
 - ▶ Should we launch a new product?
 - ▶ Should we enter a new market?
 - ▶ Should we buy another firm?
 - ▶ Should we develop a new technology?
- ▶ **Capital budgeting decisions are among the most important corporate financial decisions**
 - ▶ They reflect the firm's longer-term strategy and determine the asset side of the balance sheet as well as the firm's financing needs (liability side)
- ▶ **We need valuation tools to analyze which investment opportunities firms should undertake**

The Capital Budgeting Process

1. Find new investment projects that potentially create value for shareholders
2. Estimate the expected future cash flows
 - ▶ Determine after-tax incremental project cash flows
3. Assess project risk and the opportunity cost of capital
 - ▶ Determine a discount rate that reflects the business risk of the project and its incremental financing needs
4. Evaluate the project using a sound decision rule that
 - ▶ Selects investment that maximize shareholder value
 - ▶ Takes all cash flows into account
 - ▶ Discounts at the appropriate cost of capital Growth does not have to be a perpetuity. Firms often grow at different rates at different stages of development

Net Present Value (NPV) Rule

- ▶ The Net Present Value (NPV) of a project is the difference between the present values of its cash inflows and outflows:

$$NPV = C_0 + \frac{C_1}{1+R} + \frac{C_2}{(1+R)^2} + \dots + \frac{C_T}{(1+R)^T}$$

- ▶ The NPV Rule: Undertake an investment project if and only if its NPV is positive:

$$NPV > 0$$

- ▶ If a project's NPV is positive, then undertaking it increases firm (and hence shareholder) value

Example: The NPV Rule

- ▶ A cloud computing firm considers upgrading its data center in Texas with the latest chips. The upgrade will cost \$20 million this year and next. Once up and running, it will generate incremental free cash flows of \$30 million in two years and another \$30 million in three years, after which the new chips will fully depreciate. The firm has determined that the appropriate discount rate is 8%.
- ▶ Should the firm undertake the upgrade project?
- ▶ If the firm has 2 million shares trading at \$100, what will be the new share price when the project is announced?

Example: The NPV Rule — Answer

- ▶ **$NPV = -20 - 20 / 1.08 + 30 / 1.08^2 + 30 / 1.08^3$**
- ▶ **$= -20 - 18.52 + 25.72 + 23.81 = +\11.02 million**
- ▶ Positive: undertake the upgrade
- ▶ **The share price should rise by $11.02 / 2 = \$5.51$, from \$100 to \$105.51**
 - ▶ the market prices the project the day it is announced, not when the chips arrive
 - ▶ this is Session 8 again: the share is worth the present value of what the firm will earn

Evaluating the NPV Rule

- ▶ NPV measures how much value a project creates for the firm's shareholders
 - ▶ Since the objective of the firm is to maximize shareholder value, it should undertake all projects with positive NPV
- ▶ NPV is the right capital budgeting technique
 - ▶ It takes all cash flows into account
 - ▶ It ignores irrelevant cash flows such as sunk costs
 - ▶ It takes into account the time value of money and risk via the discount rate
- ▶ A fundamental property: NPV is additive

$$NPV \text{ of } A + B = (NPV \text{ of } A) + (NPV \text{ of } B)$$

- ▶ New projects increase firm value by the NPV of their incremental cash flows, assuming they can be estimated in isolation

Internal Rate of Return

- ▶ Notice that NPV depends on the discount rate R :

$$NPV(R) = C_0 + \frac{C_1}{1 + R} + \frac{C_2}{(1 + R)^2} + \dots + \frac{C_T}{(1 + R)^T}$$

- ▶ IRR is the discount rate at which the NPV is equal to zero:

$$NPV(IRR) = C_0 + \frac{C_1}{1 + IRR} + \frac{C_2}{(1 + IRR)^2} + \dots + \frac{C_T}{(1 + IRR)^T} = 0$$

- ▶ IRR is not a cost of capital
 - ▶ It does not incorporate market interest rates or a risk premium
 - ▶ It simply summarizes the project's cash flows on a per-dollar and per-period basis

How to Solve for IRR

- ▶ For a one-period project, IRR is just the holding period return and can be solved by hand:

$$0 = C_0 + \frac{C_1}{1 + IRR} \Rightarrow IRR = \frac{C_1}{-C_0} - 1$$

Cost of investment

- ▶ For multi-period projects, IRR is obtained numerically
 - ▶ In Excel, use “= IRR([Cash flows])”
 - ▶ What is the IRR of the data center upgrade project?



Example: IRR

- ▶ The project's expected cash flows:



- ▶ Using Excel, $IRR = 22.5\%$ solves

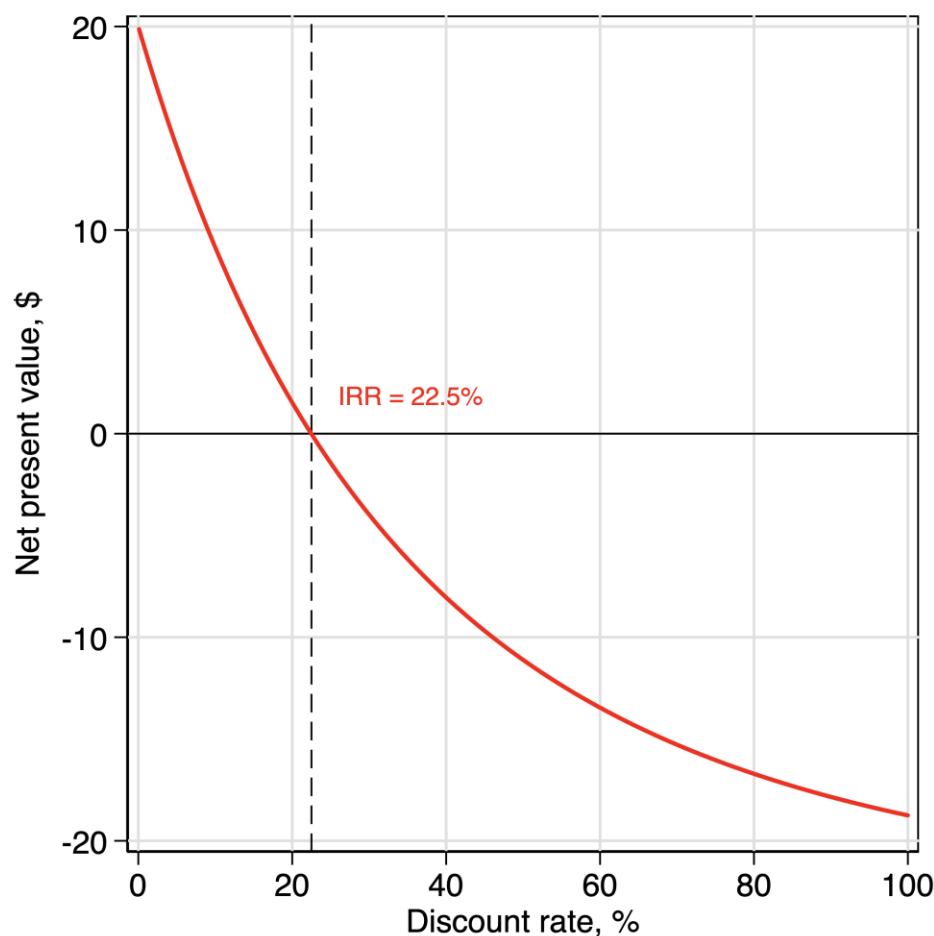
$$0 = -20 + \frac{-20}{1 + IRR} + \frac{30}{(1 + IRR)^2} + \frac{30}{(1 + IRR)^3}$$

- ▶ The IRR exceeds the project's discount rate of $R = 8\%$
 - ▶ The project is returning more than the opportunity cost of capital
 - ▶ This reflects its positive NPV

IRR Is a Project's Yield to Maturity

- ▶ In Session 6 the YTM was the rate that makes the bond's cash flows worth its price
- ▶ **The IRR is the rate that makes the project's cash flows worth its cost: $NPV(IRR) = 0$**
 - ▶ for a one-period project it is just the holding period return
- ▶ Same polynomial, same answer as before: we do not solve it by hand
 - ▶ Excel: $=IRR(\text{cash flows})$, with the cash flows in consecutive cells, starting at $t = 0$
 - ▶ for the data center: $=IRR(-20, -20, 30, 30) = 22.47\%$
- ▶ And the same caveat as the YTM: it is a summary of the cash flows, not a market rate

The IRR Rule



- ▶ IRR rule: Accept project if IRR exceeds the opportunity cost of capital

$$IRR > R$$

- ▶ Also called a “hurdle rate”
- ▶ Is the cost of capital $R < 22.5\%$?
 - ▶ If so, then $NPV > 0 \Rightarrow$ accept the project
- ▶ Is the cost of capital $R > 22.5\%$?
 - ▶ If so, then $NPV < 0 \Rightarrow$ reject
 - ▶ The firm (shareholders) are better off investing in capital markets to get R

IRR Vs NPV Rule

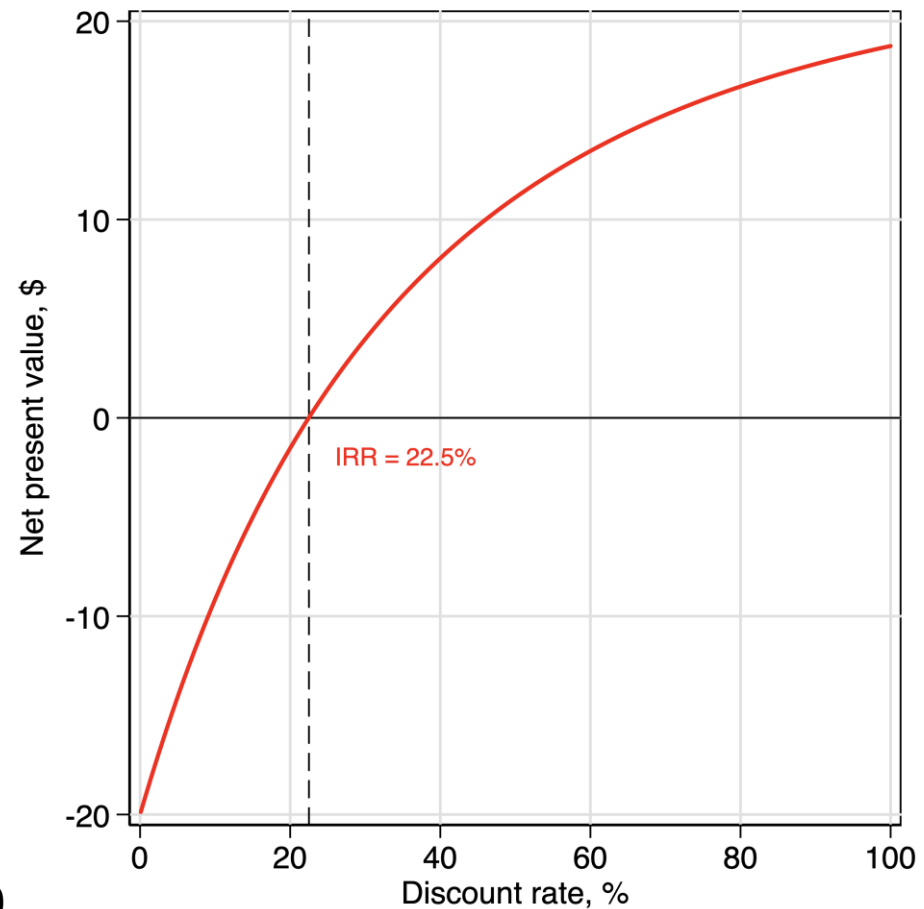
- ▶ The NPV and IRR rules lead to identical decisions when the projects have conventional cash flows and are independent
- ▶ Conventional cash flows: initial cash outflows (the initial investment) followed by cash inflows (the profits)
- ▶ Independent projects: two projects are independent if the decision to accept one does not affect the decision to accept the other

IRR Rule Pitfalls

- ▶ Do the IRR and NPV rules always coincide?
 - ▶ No. While simple and appealing, the IRR rule sometimes leads to the wrong decision (does not maximize NPV)
 - ▶ Three general case when IRR rule fails:
 1. Lending or borrowing?
 2. Multiple IRRs or IRR does not exist
 3. Mutually exclusive projects
- Violates conventional cash flows
- Violates project independence
- ▶ Let's illustrate them with examples

IRR Rule Pitfall 1: Lending or Borrowing?

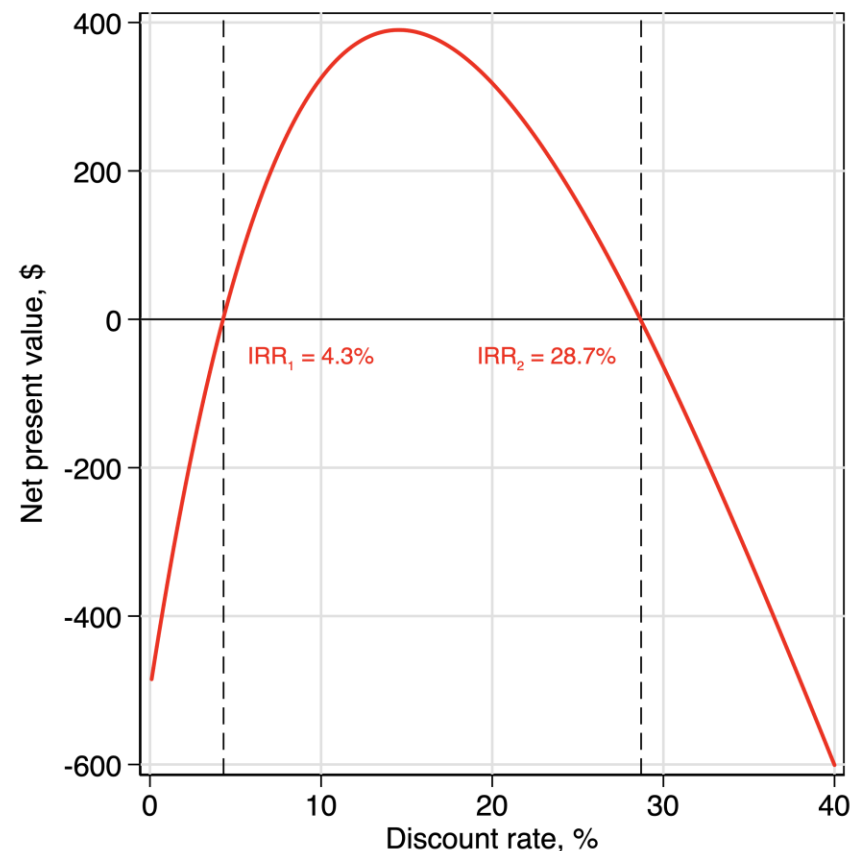
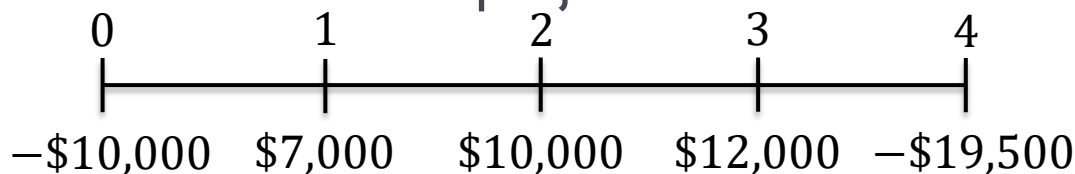
- ▶ With conventional cash flows (outflows followed by inflows), the firm is effectively lending
- ▶ When the pattern is reversed (inflows followed by outflows), it is effectively borrowing
 - ▶ The NPV is increasing in the discount rate!
 - ▶ Better to borrow at a rate lower than the cost of capital



IRR Rule Pitfall 2: Multiple IRRs

- ▶ When cash flows flip signs multiple times (outflows, followed by inflows, followed by outflows, etc.) the IRR equation can have multiple solutions
 - ▶ Intuitively, the firm is both borrowing and lending and the two dominate at different discount rates
 - ▶ Safer to use NPV rule in such cases

- ▶ Example: A large “clean-up cost” at the end of a project:



IRR Rule Pitfall 3: Mutually Exclusive Projects

- ▶ IRR is per-dollar, so it ignores the scale of a project
 - ▶ NPV is in dollars, so it takes scale into account
 - ▶ A smaller project with a higher IRR can have lower NPV than a larger project with a lower IRR:

Project	C_0	C_1	IRR	NPV at 10%
A	−\$10,000	\$20,000	100%	\$8,182
B	−\$20,000	\$35,000	75%	\$11,818

- ▶ IRR is also per period, so it ignores project length:

Project	C_0	C_1	C_2	IRR	NPV at 10%
A	−\$10,000	\$12,000	\$12,000	85%	\$10,826
B	−\$10,000	\$20,000		100%	\$8,182

- ▶ If the projects are not mutually exclusive, do both!

Evaluating the IRR Rule

- ▶ IRR as a profitability measure is easy to use and compute
- ▶ IRR gives the same answer as NPV when NPV is smoothly decreasing in the discount rate
 - ▶ This requires conventional cash flows and independent projects
- ▶ IRR is often unreliable in ranking projects of different length and scale
- ▶ Over-reliance on IRR may lead to bias in favor of shorter projects with low up-front investment

The Payback Rule

- ▶ Payback period: The number of years until the cumulative cash flows exceed the initial investment
- ▶ The Payback Rule: only accept projects that “pay back” the initial investment within a certain time frame

$$\textit{Payback period} < T$$

- ▶ The Discounted Payback Rule discounts the cash flows instead of just adding them up when computing the payback period
- ▶ The payback rule is very simple, but clearly flawed because it ignores all cash flows after the payback period
 - ▶ Leads to a bias in favor of short-term projects

Example: The Payback Rule

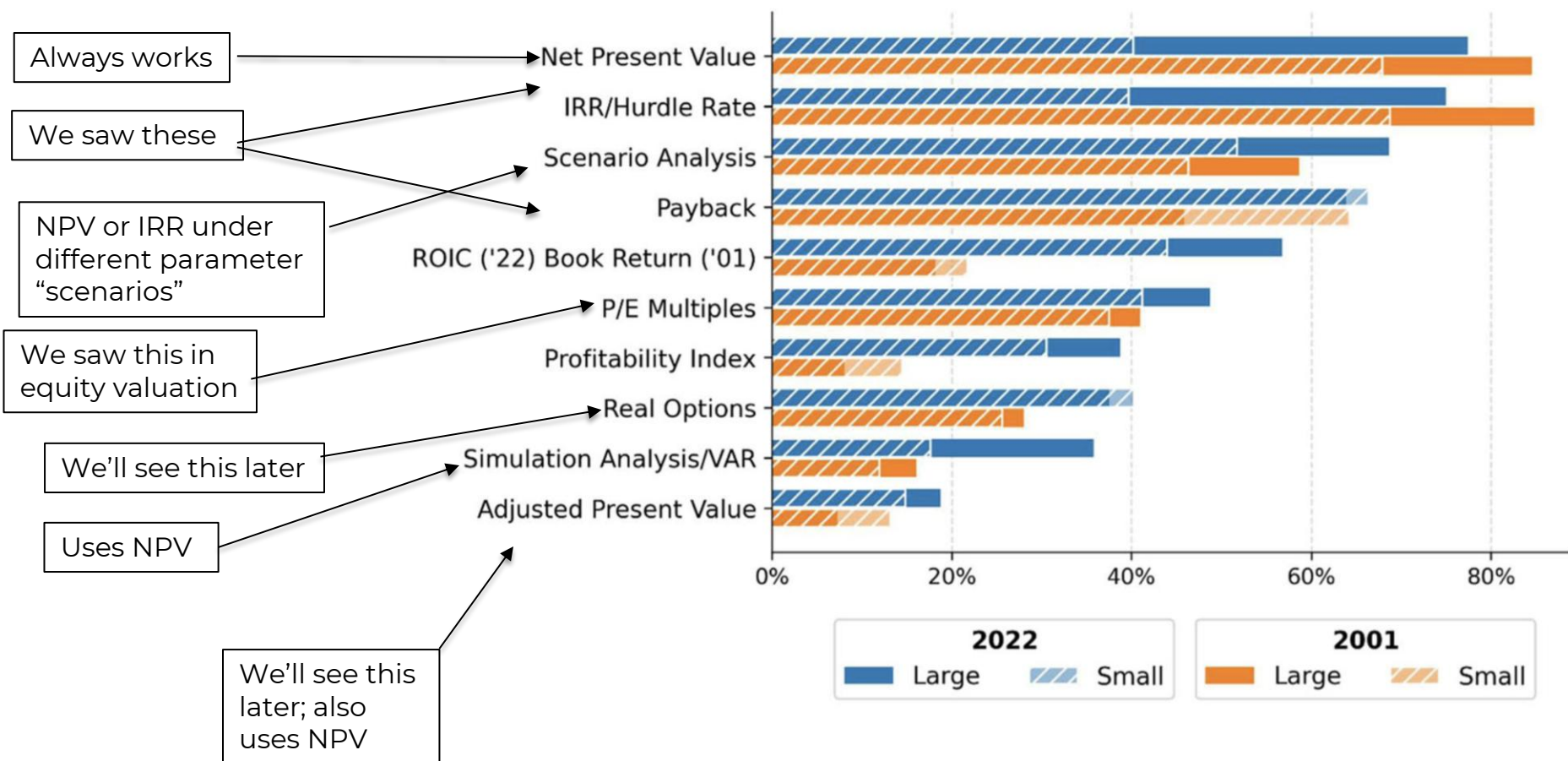
- Consider the following three projects:

	C_0	C_1	C_2	C_3	C_4	Payback
A	−\$1,000	\$800	\$400			2
B	−\$1,000	\$400	\$800			2
C	−\$1,000	\$400	\$400	\$400	\$400	3

- Calculate their discounted cash flows if $R = 12\%$:

	C_0	C_1	C_2	C_3	C_4	Discounted payback	NPV at 12%
A	−\$1,000	\$714.29	\$318.88			2	\$33.16
B	−\$1,000	\$357.14	\$637.76			—	−\$5.10
C	−\$1,000	\$357.14	\$318.88	\$284.71	\$254.21	4	\$214.94

What do Managers do in Practice?



Before the Midterm: the Sample Exams

- ▶ Two sample midterms, their solutions and the formula sheet are available on Brightspace
 - ▶ same format as the real exam: multiple choice plus numerical problems
 - ▶ they are harder than the real one, on purpose
- ▶ How to use them:
 - ▶ do one under exam conditions — formula sheet, calculator, no notes — before looking at the solutions
 - ▶ keep the other for the week of the exam
- ▶ Sessions 11 and 12, free cash flow, are on the midterm too — the samples already include them
- ▶ **Not on the midterm: duration, CAPM and beta, solving a YTM or an IRR by hand**

Summary

- ▶ Capital budgeting is valuation again: a project is worth the present value of its cash flows
- ▶ NPV rule: take every project with $NPV > 0$. It always works, and NPVs add up
- ▶ IRR is the project's yield to maturity. Take the project if $IRR > r$ — when the cash flows are conventional and the project stands alone
- ▶ IRR fails for borrowing-type cash flows, for cash flows that change sign twice, and for mutually exclusive projects of different scale or length
- ▶ Payback ignores the time value of money and everything after the cutoff
- ▶ Managers use all of them. When they disagree, follow NPV
- ▶ **Next class: Financial Statements and Free Cash Flows (BMAE 6.1–6.2)**